Answer ALL Questions

1. Question on Wind and Photovoltaic Power

a) The hourly wind speed (in m/s) profile across the year at a particular location is modelled by Weibull probability distribution where the probability of experiencing wind speed less than v_w is given by:

$$\Phi(v_w) = 1 - e^{-\left(\frac{v_w}{8}\right)^{2.2}}$$

At this location, a 8 MW direct-drive wind turbine with a permanent magnet synchronous generator (PMSG) is installed. The wind turbine parameters are given in Table 1.1 and the density of air can be assumed to be 1.22 kg/m^3.

Length of turbine blades	r	85 m
Maximum power coefficient	C_{p-max}	0.5
Optimum tip speed ratio	λ_{opt}	8
Cut-in speed	v_{w-in}	4 m/s
Cut-out speed	v_{w-out}	25 m/s

Table 1.1: Parameters of the wind turbine

i) Calculate the rated wind speed (in m/s)

[1]

$$P_w = \frac{1}{2}\rho A v_w^3 C_{p-max}$$

$$\Rightarrow v_R = \left(\frac{2P_{rated}}{\rho\pi r^2 C_{p-max}}\right)^{\frac{1}{3}} = 10.49 \text{ m/s}$$

ii) What is the probability of encountering wind speeds higher than the rated speed?

[1]

$$p = 1 - \Phi(10.49) = 12.32\%$$

iii) What is the probability of encountering wind speeds in the region where the wind turbine operates in maximum power extraction mode by adjusting the turbine rotational speed?

[1]

$$p = \Phi(10.49) - \Phi(4) = 65.49\%$$

iv) Determine the range of rotational speed (in rpm) of the turbine required to extract maximum power from wind.

Solution

[2]

$$\text{TSR } \lambda = \frac{\omega_T \times r}{v_w} \Rightarrow \omega_{T-opt} = \frac{\lambda_{opt} \times v_w}{r}$$

[1 mark]

$$\left(\frac{\lambda_{opt} \times v_{w-cutin}}{r} \times \frac{60}{2\pi}\right) \leq \omega_{T-opt} \leq \left(\frac{\lambda_{opt} \times v_R}{r} \times \frac{60}{2\pi}\right)$$

$$\Rightarrow 3.6 \text{ rpm} \leq \omega_{T-opt} \leq 9.4 \text{ rpm}$$

[1 mark]

v) How many poles are needed for the PMSG if the minimum allowable stator frequency is 10 Hz?

[2]

Synchronous speed (in rpm)

$$N_s = \frac{60 \times f}{p/_2} \Rightarrow p = \frac{120 \times f}{N_s}$$

[1 mark]

For

$$f \geq 10 \text{ Hz}, p \geq \frac{120 \times f}{N_{s-min}} = 333.79$$

Number of poles 334

[1 mark]

b) The declination angle of earth on a particular day is 18°. For this particular day and a specific location with a latitude of 50°N:

i) Calculate the elevation angle (in degrees) and the air mass (AM) at solar noon.

[2]

$$\alpha = 90° - (\phi - \delta) = 58°$$

[1 mark]

$$AM = \frac{1}{\sin \alpha} = 1.179$$

[1 mark]

ii) Determine the optimum tilt angle (in degrees with respect to the horizontal) of a solar panel at solar noon in order to maximise the direct irradiance.

[2]

Solution

Angle of incidence on a solar panel tilted by an angle β is

$\alpha + \beta = 90° - (\phi - \delta) + \beta$
[1 mark]

Irradiance on solar panel is maximum when the angle of incidence is 90°
$90° - (\phi - \delta) + \beta_{opt} = 90° \Rightarrow \beta_{opt} = \phi - \delta = 32°$
[1 mark]

b) Partial shading of PV modules or PV arrays are known to be problematic.

i) State two problems associated with shading of a few PV cells within a PV module with series connected PV cells.

[2]

- The current flowing through the PV module is limited by that pf the shaded PV cell(s) which could be very low depending on the extent of shading
- The shaded PV cell(s) develops a negative voltage and dissipates power which heats those up.

[1 mark for each point]

ii) Mention the role of string diodes in a solar PV array with parallel strings of PV modules and state the drawback of using the string diodes.

[2]

- A string of PV cells/modules with shaded/warmer cell(s) have reduced voltage resulting in reverse current flow and further heating. String diodes used to block reverse current.
- String diodes prevent the affected strings from delivering any current i.e. power.

[1 mark for each point]

d) A solar PV module with 96 PV cells has a Watt Peak rating of 350 W corresponding to the standard test condition (STC). The PV cells are connected in series with a bypass diode across groups of 32 of them. The open circuit voltage and short circuit current of each cell under STC is 0.7 V and 6.5 A, respectively. The efficiency of the PV cells is 16.5% under STC. A PV array is formed with 3 parallel strings. Each string has 4 PV modules in series with a string diode. The total current from the 3 strings is 18 A for maximum power output at STC.

Neglecting other losses within the PV module apart from that in the PV cells, calculate the following assuming STC for all the PV cells:

Solution

i) Fill factor of each PV cell.

[1]

Maximum power for each cell

$$P_{max-cell} = \frac{P_{max}}{N_{cell}} = 3.646\ \text{W}$$

Fill factor is

$$\text{FF} = \frac{P_{max-cell}}{V_{OC} I_{SC}} = 0.8$$

ii) Surface area of each PV cell in cm^2.

[1]

$$\eta = \frac{P_{max-cell}}{P_{in}} = \frac{P_{max-cell}}{H_{STC} A_{cell}}$$

$$\Rightarrow A_{cell} = \frac{P_{max-cell}}{P_{in}} = \frac{P_{max-cell}}{\eta H_{STC}} \times 10^4 = 243.06\ \text{cm}^2$$

iii) Voltage at which the PV cells produce maximum power.

[1]

PV cell current at maximum power point

$$I_{max-cell} = \frac{I_{arr}}{N_{str}} = 6\ \text{W}$$

$$V_{max-cell} = \frac{P_{max-cell}}{I_{max-cell}} = 0.6076\ \text{V}$$

iv) Power (in kW) generated by the PV array if one PV module within a particular string is completely shaded.

[2]

With one PV module completely shaded and bypassed by the bypass diode, the voltage of that particular string will be less than the other two. Hence, the string diode of that string won't conduct reducing the current and hence the power output of the PV array to two-third of the rated power.

[1 mark]

$$P = \frac{2}{3} \times P_{R-arr} = \frac{2}{3} \times P_{R-mod} \times N_{mod} = 2.8\ \text{kW}$$

[1 mark]

Solution

Solution

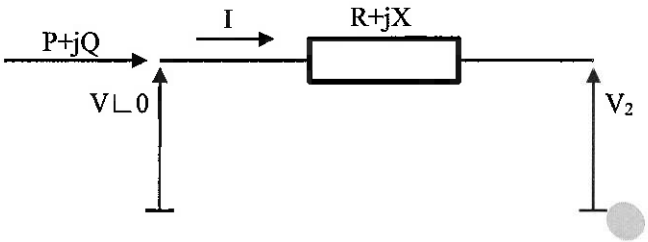
2. Question on Distributed Generation, Voltage Control and Security of Supply

Solution:

Part A

a)

Figure shows equivalent diagram of connection



Voltage across R and X is given by:

$$\overrightarrow{\Delta V} = \sqrt{3} \cdot I \cdot (R + jX) = \sqrt{3} \cdot \frac{P - jQ}{\sqrt{3} \cdot V} \cdot (R + jX)$$

Given that for voltage V, angle is assumed zero the conjugate complex value is also V.
Multiplying apparent power and impedance gives:

$$\overrightarrow{\Delta V} = \frac{P \cdot R + Q \cdot X + j(P \cdot X - Q \cdot R)}{V}$$

The voltage rise is approximately given by the formula:

$$\Delta V = |\vec{V}| - |\vec{V_2}| \approx \frac{R \cdot P + X \cdot Q}{V}$$

It should be noted that typically:

$$\Delta V = |\vec{V}| - |\vec{V_2}| \neq |\overrightarrow{\Delta V}| = |\vec{V} - \vec{V_2}|$$

In the case of 11kV connection, maximum allowed voltage at busbar C is 1.06 and the formula could be written as:

$$\Delta V = V_C - V_A \approx \frac{r \cdot L \cdot P + x \cdot L \cdot P \cdot \tan(\cos^{-1} 0.96)}{V_C}$$

$$\mathrm{P} = \frac{V_C \cdot (V_C - V_A)}{L \cdot (r + x \cdot \tan(\cos^{-1} 0.96))}$$

$$\mathrm{P_{max}} = \frac{1.06 \cdot 11 \cdot (1.06 \cdot 11 - 1.03 \cdot 11)}{6 \cdot (0.07 + 0.07 \cdot \tan(\cos^{-1} 0.96))} = 7.1\ MW$$

Rating of the cable is greater than the installed wind farm capacity

$$S_r = \sqrt{3} \cdot 11 \cdot \frac{900}{1000} = 17.1\ MVA > \frac{7.1}{0.96} = 7.4\ MVA$$

Maximum capacity of wind farm connection on 11 kV network at fixed OLTC is 7.1 MW.

Solution

The minimum voltage at busbar A while keeping voltage limits at busbar B is:

$$V_{Amin} \approx V_{Bmin} + \frac{L \cdot P \cdot (r + x \cdot \tan(\cos^{-1} 0.96)}{V_{Bmin}}$$

$$V_{Amax} \approx 0.94 + \frac{4 \cdot 20 \cdot (0.07 + 0.07 \cdot \tan(\cos^{-1} 0.95))}{0.94 \cdot 11^2} = 1.0036\ p.u.$$

Maximum capacity of wind farm connection on 11 kV network at lowest possible voltage at busbar A is:

$$P_{max2} = \frac{1.06 \cdot 11 \cdot (1.06 \cdot 11 - 1.0036 \cdot 11)}{6 \cdot (0.07 + 0.07 \cdot \tan(\cos^{-1} 0.96))} = 20.0\ MW$$

$$S_{max2} = \frac{P_{max2}}{0.96} = \frac{20.0}{0.96} = 20.8\ MVA$$

However, the selected cable can only support connection of up to 17.1 MVA wind farm. Maximum capacity of wind farm connection is:

$$P_{max2} = 0.96 \cdot S_r = 0.96 \cdot 17.1 = 16.5\ MW$$

Rating of the cable is not adequate for connection of greater wind capacity.

b)

Distributed generation hosting capacity is driven by thermal (asset capacity or security clearance of overhead lines), voltage and fault level constraints. Area voltage control can increase distribution generation hosting capacity by actively managing voltage at its connection busbar such that voltages at different points in the network, depending on generation and demand on different feeders, are within limits.

Part B:

a)

Distributed generation exports electricity within group demand area reducing upstream network capacity need during a network asset outage.

Electricity storage could discharge during the network congestion period (during a network asset outage).

Demand side management could switch-off or postpone non-essential part of its demand, change the primary source of energy e.g. hybrid heat pump could use gas instead of electricity, and utilise stored energy e.g. vehicle-to-grid approach to reduce need for electricity during network congestion period.

Dynamic line and transformer rating could release "additional network capacity" by more accurately estimating the asset emergency rating.

b)

Capacity outage probability table of three identical generating units is

Solution

Capacity in	Probability
3C	A^3
2C	$3\ A^2\ (1\text{-}A)$
C	$3\ A\ (1\text{-}A)^2$
0	$(1\text{-}A)^3$

Expected energy not supplied (EENS) for three generating units, during peak season, when peak demand is equal to 30 MW, during peak season is

Capacity in, MW	Probability	EENS, p.u.
30	$0.86^3 = 0.6361$	0
20	$3\ 0.86^2\ (1\text{-}0.86) = 0.3106$	$0.3106\ (10\ 0.05) = 0.1553$
10	$3\ 0.86\ (1\text{-}0.86)^2 = 0.0506$	$0.0506\ (20\ 0.05 + 8\ 0.35) = 0.1922$
0	$(1\text{-}0.86)^3 = 0.0027$	$0.0027\ (30\ 0.05 + 18\ 0.35) = 0.0214$
Total	**1**	**0.3689**

Alternative calculation of EENS when peak period duration is scaled to 1 is also acceptable.

Firm capacity which have the same EENS is

$$FC = 30 - \frac{EENS}{0.05} = 30 - \frac{0.3689}{0.05} = 22.62\ MW$$

Plant contribution during peak season is

$$C = \frac{FC}{PC} = \frac{22.62}{30} = 75\%$$

Capacity outage probability table of two identical generating units is

Capacity in	Probability
2C	A^2
C	$2\ A\ (1\text{-}A)$
0	$(1\text{-}A)^2$

Expected energy not supplied (EENS) for two generating units, during off-peak season, is

Capacity in, MW	Probability	EENS, p.u.
20	$0.86^2 = 0.7396$	$0.7396\ (4\ 0.2) = 0.5917$
10	$2\ 0.86\ (1\text{-}0.86) = 0.2408$	$0.2408\ (14\ 0.2 + 2\ 0.4) = 0.8669$
0	$(1\text{-}0.86)^2 = 0.0196$	$0.0196\ (24\ 0.2 + 12\ 0.4) = 0.1882$
Total	**1**	**1.6467**

Firm capacity which have the same EENS is

$$FC_2 = 24 - \frac{EENS}{0.2} = 24 - \frac{1.6467}{0.2} = 15.77 MW$$

Solution

Plant contribution during off-peak season is

$$C_2 = \frac{FC_2}{PC} = \frac{15.77}{30} = 53\%$$

Plant contribution is lower during off-peak season given that only two units operate due to one unit being in maintenance.

Solution

3. Question on Low Carbon Heat

a)

Building internal set temperature, K
External temperature, K
Other weather factors, e.g. solar gain, wind, K
Fabric (walls, windows, etc) thermal transmittance, W/m^{2}K
Air changes or permeability, m3/m2.h @ 50Pa specified pressure
Other heat sources, e.g. occupants, appliances, etc

b)

i) From the graph:

Space and water heat demand is 1.75TWh/d
Hence space heat demand is 1.75 – 0.25 = 1.5TWh/d
Average building space heat demand is (1.5TWh/d x 200 days)/30 million = 10MWh

ii) From the graph: Water heat demand is (0.25TWh/d * 365)/30million = 3.04MWh

c) Simplified expression of space heat demand: Space heat for the period = (Average internal and external temperature difference) x heating period in hours x average building floor area x HLP
So re-arranging and substituting: HLP = 10MWh/[(17°C-7°C)x100m^{2}x200daysx24 hours] = 2.08W/m^{2}K

c)

Climate change is estimated to result in an increase in average temperatures of 2°C for the heating period by 2050. Assuming a 30% average improvement in building insulation, determine

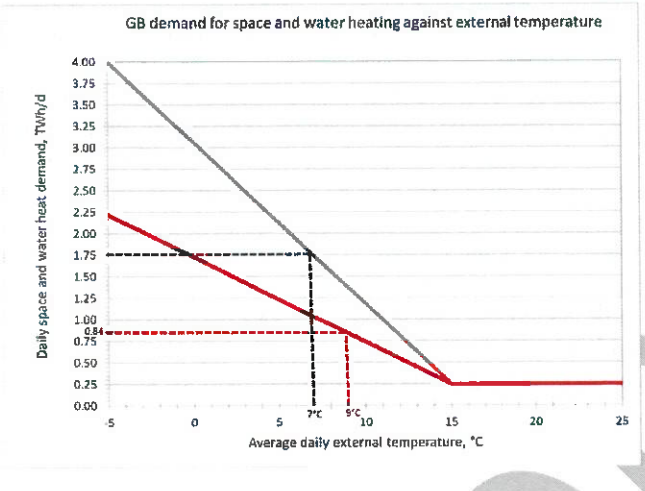
i) Substituting in the equation in c) above:
Space heat demand = (17°C - 9°C) x 100m^2 x 200days x 24 x 2.08 (100% - 30%) = 5.6MWh

ii) Total GB demand for domestic space and water heating: (5.6MWh + 3.04 MWh) x 30 million = 259.2TWh

d)

Total space heat demand at 9°C is (5.6MWh x 30 million)/200days = 0.84 TWh/d
Assuming: No change in number of buildings, No change in heating period, No change in building internal temperature, No change in water consumption

Solution



e)

ASHP transfers thermal energy from the outside air to the area to be heated. There are 4 stages to the cycle and these comprise:

- Evaporation - The refrigerant passes through a heat exchanger where it evaporates absorbing heat in the process.
- Compression - The heated refrigerant is then compressed to higher pressure using an electrically driven compressor.
- Condensation - The compressed refrigerant passes through another heat exchanger located in the space to be heated and condenses, changing state back to a liquid and rejecting heat in the process.
- Expansion – The liquid refrigerant passes through an expansion device lowering its pressure prior to returning to the evaporator.

f)

i) If the average efficiency of an air source heat pump is 250%, then total electricity demand increase is:

259.2 TWh/250% = 104TWh

ii) Daily demand at -5°C is 2.25TWh. As the average daily external temperature falls the air source heat pump must do more work to raise the temperature of the air to allow the desired internal temperature to be reached and maintain the building heat requirements. Hence there must be more electrical input for the same heat output and so the efficiency of the heat pump falls.

iii) If the efficiency of the heat pump at -5°C is 200% then the average building electricity demand for the day is: 2.25TWh/(14Mx200%) = 80.4kWh

Solution

4. Question on Wind Balancing, Hydro Power and Offshore Transmission

a) Uncertainty in wind forecast becomes a source of additional balancing requirements which can be fairly substantial in magnitude. Wind forecast time scales are important for determining reserve requirements. For time scales from several seconds to a few minutes, the fluctuation of the overall output of wind generation will be small given that there is considerable diversity in outputs of individual wind farms. In these very short (Response) time scales, the dominant variability factor is the potential loss of conventional plant, not fluctuations in wind power. Reserve requirements are concerned with the wind forecast uncertainty over longer time scales of minutes to hours. Wind forecast techniques vary for different time scales. For longer horizons beyond several hours, forecasts based on meteorological information are preferred. For short-term forecasts, up to several hours ahead, such methods are out-performed by various statistical techniques.

b)

i) 18 CCGTs will be required to run to provide the reserve required (200x18=3,600)

ii) Cost associated with reserve 43,200 £/h =0.2x60x3600

iii) Additional CO2 emitted (due to the need to provide reserve) 302.4 t/h = 0.2x0.42x3600

iv) Amount of wind power that may need to be curtailed at the minimum demand condition:

Nuclear output (GW)	5
CCGT output for reserve (GW)	5.4 (=18x300)
Net demand to be served by wind (GW)	7.6 (=18 – 10.4)
Wind power curtailed (GW)	4.4 = 12-7.6

No curtailment during peak demand condition

Hydro Power

c) Each unit of a reservoir based hydro power plant has a rated power capacity of 800 MW. The heights of different parts of the plant from a reference level are given in Table 4.1:

Surface of headwater in reservoir	h_1	110 m
Inlet point into penstock	h_2	30 m
Turbine inlet and outlet	$h_3 = h_4$	0 m
Surface of tailwater in river below	h_5	10 m

Table 4.1: Heights of different parts of the hydro power plant from a reference level

The penstock has a constant inner diameter of 12 m. The combined efficiency of the turbine generator unit is 90%. The density of water is 1000 kg/m^3 and acceleration due to gravity is 9.81 m/s^2. Atmospheric pressure is 101.32 kPa. Neglect any loss of head.

i) How much volume flow rate or discharge (in m^3/s) of water is required for rated power output from the turbine generator unit?

Solution

[2]

Neglecting any loss of head, available head

$H = h_1 - h_2 = 100$ m
[1 mark]

$P = \rho g Q H \eta$

$\Rightarrow Q = \frac{P}{\rho g H \eta} = 906.1\ \mathrm{m}^3/s$
[1 mark]

ii) What is the velocity (m/s) and absolute pressure (kPa) of water at the inlet of the turbine?

[3]

Cross-sectional area of the penstock

$A_p = \pi R^2 = 113.1\ \mathrm{m}^2$

Velocity of water at penstock and turbine inlet

$v_2 = v_3 = \frac{Q}{A_p} = 8.01$ m/s
[1 mark]

Using Bernoulli's principle

$p_{atm} + \rho g h_1 = p_2 + \rho g h_2 + \frac{1}{2}\rho v_2^2 = p_3 + \rho g h_3 + \frac{1}{2}\rho v_3^2$
[1 mark]

$\Rightarrow p_3 = p_{atm} + \rho g (h_1 - h_3) - \frac{1}{2}\rho v_3^2 = 11483$ kPa
[1 mark]

Offshore Transmission

d) Justify the following statements giving appropriate reasons:

i) An LCC HVDC substation requires larger space than an equivalent VSC HVDC substation.

[2]

- For a LCC HVDC substation, capacitor banks are needed to provide the reactive power demanded by the converters. Filters are required to suppress the low-order current (AC-side) and voltage harmonics (DC-side). Special type of transformer arrangement is also necessary. All these contribute to a relatively large space requirement for LCC HVDC substation.

- For a VSC HVDC substation, no reactive power provision is necessary and filtering requirement is very little or zero (with present technology). Special

Solution

transformer arrangement is not required. Hence, the space required a VSC HVDC substation is much less compared to an LCC HVDC substation.

[1 mark for each point]

ii) Stronger and lighter polymeric cables can be used for VSC HVDC links but not for LCC HVDC links.

[2]

- For an LCC HVDC link, power flow is reversed by flipping the direct voltage polarity at both ends as current direction through the HVDC link cannot be reversed. For a VSC HVDC link, direct voltage polarity is fixed, and power is reversed by changing the direction of current flow through the HVDC link.

- Polymeric cables cannot withstand reversal of voltage polarity from time to time and hence can only be used with VSC HVDC but not LCC HVDC.

[1 mark for each point]

iii) For sub-sea cable transmission, the breakeven distance beyond which HVDC is more cost-effective than AC is larger in case of a remote offshore wind farm connection than a sub-sea interconnector between two onshore AC systems.

[1]

Breakeven distance depends on the fixed cost (i.e. cost of terminal components which is incurred even for zero/small distance) and the increasing cost (of cables etc.) with distance. For remote offshore wind farms, the fixed cost is higher due to much more expensive offshore platform than AC which is not needed for an interconnector. The increasing cost is similar in both cases resulting in a larger breakeven distance for remote wind farm connection.

Solution